

Discriminating Batch Artifacts from Biological Signal in Metagenomic Sequences via Tandem Repeat Entropy

Andrew Scouten^{1,*}

¹Computer Science Department, Texas State University, 601 University Drive, 78666, TX, USA

Abstract

TODO: Abstract
See below ⇒

Keywords metagenomics, tandem repeats, contamination, batch effects, cancer microbiome

1. Problem Statement & Hypothesis

Metagenomic datasets are heavily corrupted by systematic noise injected during sample processing: reagent- and kit-derived contaminant taxa (Salter et al., 2014). This presents a fundamental data provenance problem. Biologically, genuine host-associated microbes exist as diverse, evolving populations with fine strain structure across individuals. Conversely, bacteria that sneak in from lab kits or chemicals will look almost exactly the same across all patients because they are exact copies of a few specific strains that contaminated the manufacturing process.

Because tandem repeat (TR) loci are the fastest-evolving sites in bacterial genomes (van Belkum et al., 1998), they serve as high-resolution hashes or fingerprints to track sequence provenance. By shifting from TR characterization to artifact detection, we can convert a biological measurement into a computational filter. De novo contaminant detection without negative controls has previously been approached at the level of cross-sample taxonomic co-occurrence (Liu et al., 2022); the question here is whether within-taxon sequence variation supplies a signal at finer resolution.

H1 (primary).

True biological signal will exhibit high cross-sample entropy in TR allele lengths at a given locus. Batch artifacts (contaminants) will exhibit low entropy across samples. This entropy gap should be mathematically detectable even at the low sequencing depths characteristic of tumor samples.

H2 (secondary).

Since contaminants are lab-derived artifacts, their TR signatures will cluster according to where and when they were sequenced (by center and batch). Genuine biological TR calls will not group by these technical variables.

2. Data Architecture

The project relies on the sources in Table 1.

3. Computational Pipeline & Open-Source Integration

This pipeline will directly leverage and extend the existing TR finding architecture in the dashnowlab/inTRuder repository (Dashnow et al., 2026) of which I am a contributor (from a hackathon). Ongoing software engineering contributions to inTRuder will be made concurrently with the research to improve microbial TR locus discovery for the broader bioinformatics community.

- Reference Cataloging & Alignment:** Construct a baseline index of TR loci using inTRuder (Dashnow et al., 2026) and cataloging logic adapted from Phasomelt (Aidley et al., 2018).
- Allele Calling & Error Correction:** Anchor reads to the cataloged loci and measure allele lengths, applying a probabilistic stutter model adapted from human STR genotyping (Dashnow et al., 2022; Willems et al., 2017) to correct for short-read sequencing errors.
- Confounder Control:** Calculate cross-patient TR entropy. Potentially use a bias-corrected entropy estimator (e.g. Chao–Shen; Chao and Shen, 2003) with sequencing depth included as a model covariate. Explicitly avoid downsampling (rarefying) the data, as plug-in estimates of alpha diversity are negatively biased and rarefaction artificially exacerbates this bias at low depths (Willis, 2019).
- Classification & Validation:** Train a gradient boosting model with post-hoc SHAP attribution (Chen and Guestrin, 2016; Lundberg and Lee, 2017) using these TR diversity and batch metrics to classify sequences. Validate the algorithm's performance against the standard decontam method as a baseline comparator (Davis et al., 2018), and confirm

Table 1. Data sources and their role in the pipeline.

Data Source	Pipeline Role
HMP / iHMP (NIH Common Fund) (Human Microbiome Project Consortium, 2012; Integrative HMP (iHMP) Research Network Consortium, 2019)	<i>Positive Control:</i> High-depth oral and gut data. Used to establish the baseline entropy of TR alterations for taxa shared with tumor datasets (e.g. <i>Fusobacterium</i> , <i>Streptococcus</i> , oral anaerobes).
TCGA COAD and STAD (NCI GDC) (Grossman et al., 2016)	<i>Target Domain (Stress Test):</i> The contested dataset where original tumor microbiome signals were subsequently retracted (Poore et al., 2024). Requires dbGaP access for raw files.
Published Decontamination Reanalyses	<i>Evaluation Labels:</i> Retained and rejected taxon calls from an independent reanalysis (Gihawi et al., 2023). The errors documented there are database-driven misclassification of human reads and a normalization artifact rather than reagent clonality, so these labels test whether the TR signal transfers to an adjacent failure mode rather than validating the mechanism proposed in H1 directly. The decontam package (Davis et al., 2018) is held out as the baseline comparator rather than used as a source of labels.

it successfully flags the specific analytical failure modes documented by Gihawi et al. (2023).

References

J. Aidley, J. J. Wanford, L. R. Green, S. K. Sheppard, and C. D. Bayliss. Phasomelt: an 'omics' approach to cataloguing the potential breadth of phase variation in the genus *Campylobacter*. *Microbial Genomics*, 4(11), 2018. doi: 10.1099/mgen.0.000228.

A. Chao and T.-J. Shen. Nonparametric estimation of Shannon's index of diversity when there are unseen species in sample. *Environmental and Ecological Statistics*, 10(4): 429–443, 2003.

T. Chen and C. Guestrin. XGBoost: a scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.

H. Dashnow, B. S. Pedersen, L. Hiatt, et al. STRling: a k-mer counting approach that detects short tandem repeat expansions at known and novel loci. *Genome Biology*, 23: 257, 2022.

H. Dashnow, A. K. Avvaru, B. Jadhav, et al. inTRuder: identifying novel TR loci and motifs in individuals that are not in the reference genome. <https://github.com/dashnowlab/inTRuder>, 2026. Fork of <https://github.com/collaborativebioinformatics/inTRuder>. MIT licence. Accessed 13 September 2026; commit TODO.

N. M. Davis, D. M. Proctor, S. P. Holmes, D. A. Relman, and B. J. Callahan. Simple statistical identification and removal of contaminant sequences in marker-gene and metagenomics data. *Microbiome*, 6:226, 2018. doi: 10.1186/s40168-018-0605-2.

A. Gihawi, Y. Ge, J. Lu, et al. Major data analysis errors invalidate cancer microbiome findings. *mbio*, 14(5):e01607–23, 2023. doi: 10.1128/mbio.01607-23.

R. L. Grossman, A. P. Heath, V. Ferretti, et al. Toward a shared vision for cancer genomic data. *New England*

Journal of Medicine, 375(12):1109–1112, 2016. doi: 10.1056/NEJMp1607591.

Human Microbiome Project Consortium. Structure, function and diversity of the healthy human microbiome. *Nature*, 486: 207–214, 2012. doi: 10.1038/nature11234.

Integrative HMP (iHMP) Research Network Consortium. The Integrative Human Microbiome Project. *Nature*, 569:641–648, 2019. doi: 10.1038/s41586-019-1238-8.

Y. Liu, R. A. L. Elworth, M. D. Jochum, K. M. Aagaard, and T. J. Treangen. De novo identification of microbial contaminants in low microbial biomass microbiomes with Squeeze. *Nature Communications*, 13:6799, 2022. doi: 10.1038/s41467-022-34409-z.

S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*, pages 4765–4774, 2017.

G. D. Poore, E. Kopylova, Q. Zhu, et al. Retraction note: Microbiome analyses of blood and tissues suggest cancer diagnostic approach. *Nature*, 631:694, 2024. doi: 10.1038/s41586-024-07656-x.

S. J. Salter, M. J. Cox, E. M. Turek, et al. Reagent and laboratory contamination can critically impact sequence-based microbiome analyses. *BMC Biology*, 12:87, 2014. doi: 10.1186/s12915-014-0087-z.

A. van Belkum, S. Scherer, L. van Alphen, and H. Verbrugh. Short-sequence DNA repeats in prokaryotic genomes. *Microbiology and Molecular Biology Reviews*, 62(2):275–293, 1998.

A. J. Vogler, C. Keys, Y. Nemoto, R. E. Colman, Z. Jay, and P. Keim. Effect of repeat copy number on variable-number tandem repeat mutations in *Escherichia coli* O157:H7. *Journal of Bacteriology*, 188(12):4253–4263, 2006. doi: 10.1128/JB.00001-06.

T. Willems, D. Zielinski, J. Yuan, A. Gordon, M. Gymrek, and Y. Erlich. Genome-wide profiling of heritable and de novo STR variations. *Nature Methods*, 14(6):590–592, 2017. doi: 10.1038/nmeth.4267.

A. D. Willis. Rarefaction, alpha diversity, and statistics. *Frontiers in Microbiology*, 10:2407, 2019. doi: 10.3389/

fmich.2019.02407.

M. Zepeda-Rivera, S. S. Minot, H. Bouzek, et al. A distinct *Fusobacterium nucleatum* clade dominates the colorectal cancer niche. *Nature*, 628(8007):424–432, 2024. doi: 10.1038/s41586-024-07182-w.